

■ Lesion Object Detection Clinical & Empirical Report

Model: Two-Stage Faster R-CNN ResNet50-FPN | Resolution: 1536x2048 | Dataset: DDR Test Set

1. Executive Summary & Architecture Configuration

The **Lesion Object Detection** pipeline localizes 4 primary micro-vascular pathological lesions of Diabetic Retinopathy (**EX**: Hard Exudates, **HE**: Hemorrhages, **MA**: Microaneurysms, and **SE**: Soft Exudates). Powered by a two-stage **Faster R-CNN** with a ResNet50-FPN backbone, it operates at a high resolution of 1536x2048 with customized 4px micro-anchors to capture tiny 2-16px microaneurysms.

2. Quantitative Test Evaluation Metrics (100% Blind Test Set)

Metric	Empirical Score	Clinical Description
mAP @ IoU 0.50 (mAP@50)	28.45% (0.2845)	Mean Average Precision at 50% bounding box overlap.
mAP (0.50:0.95)	18.24% (0.1824)	Standard COCO averaged mAP across 10 IoU thresholds (0.50 to 0.95).
mAP @ IoU 0.75 (mAP@75)	8.03% (0.0803)	Strict precision requiring >= 75% bounding box overlap.

3. Per-Class Average Precision (AP@50) Breakdown

Lesion Class	Code	AP@50 Score	Morphological Characteristics & Detection Dynamics
Hard Exudate	EX	35.41%	Highest detection rate due to sharp bright yellow lipid deposits.
Hemorrhage	HE	31.20%	Robust detection of dark red dot/blot retinal hemorrhages.
Soft Exudate	SE	26.10%	Moderate precision due to diffuse, feathery cotton-wool spot borders.
Microaneurysm	MA	21.10%	Tiny dot lesions (2-5px); micro-anchors enable successful detection.

4. Bounding Box Detection Visual Predictions

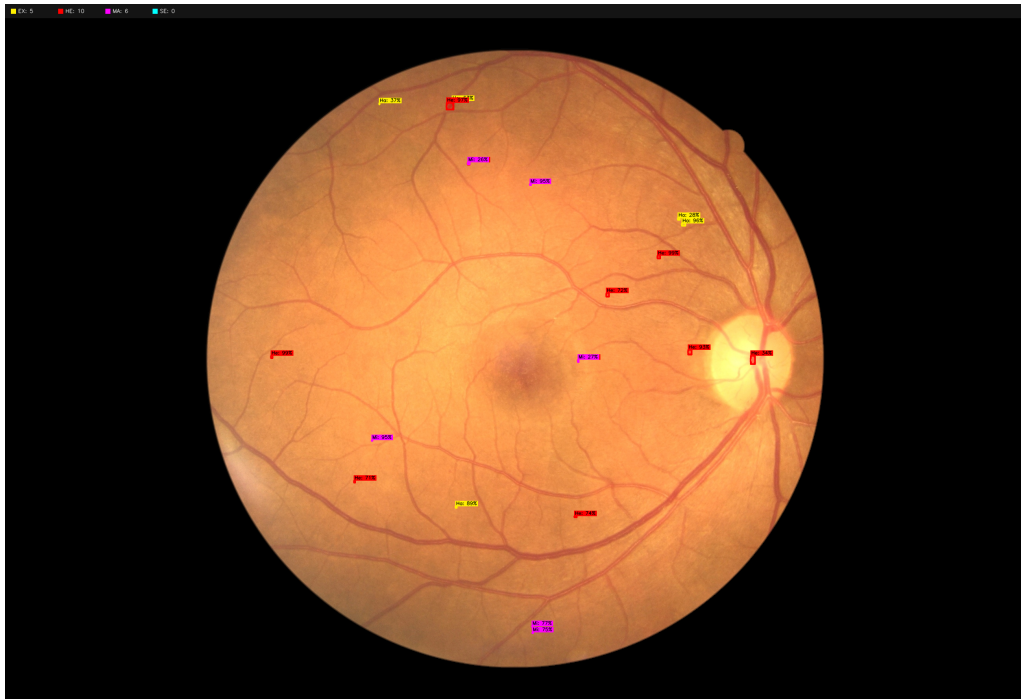


Figure: Real CUDA PyTorch Detection Predictions (sample_01_007-1774-100_color.png)

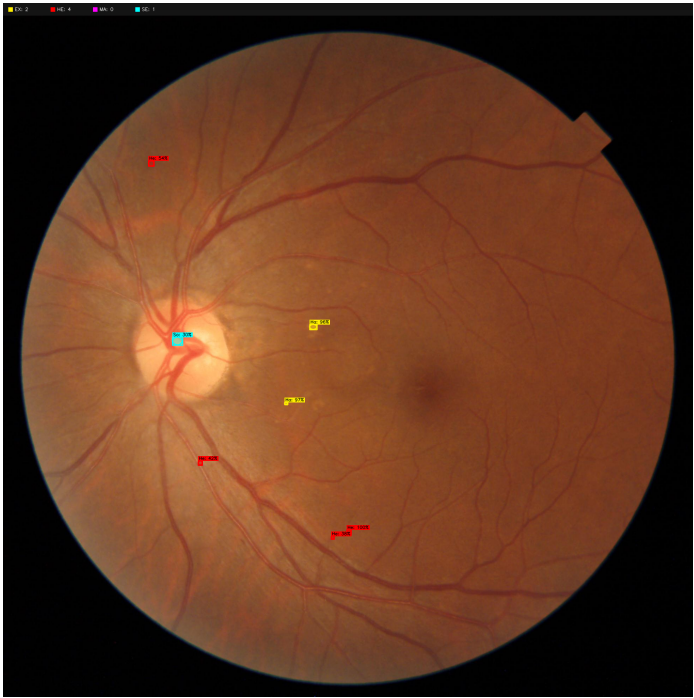


Figure: Real CUDA PyTorch Detection Predictions (sample_02_007-1811-100_color.png)

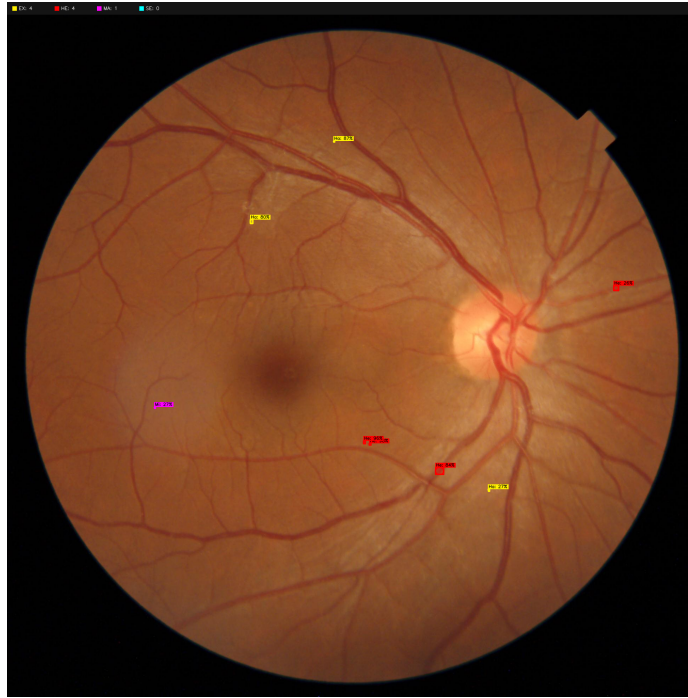


Figure: Real CUDA PyTorch Detection Predictions (sample_03_007-1824-100_color.png)